

ECE 3340 Numerical Methods

Homework 3: Complexity Analysis

Name: |

ID: |

Solve the following problems from **Chapter 3, Discrete Math and Complexity**. Use any available space to work out the problem and **place your final solution in the box provided**.

Problem 1: Provide the computational complexity of each operation using big-O notation.

$$O(N^2)$$

$\mathbf{A}\mathbf{v}$ where $\mathbf{A} \in \mathbb{R}^{N \times N}$

$$O(N)$$

Multiply N vectors $\mathbf{v}_i \in \mathbb{R}^3$ by a transformation matrix $T \in \mathbb{R}^{3 \times 3}$

$$O(MN)$$

$\mathbf{Y} = \mathbf{A}\mathbf{B}\mathbf{x}$ where $\mathbf{A} \in \mathbb{R}^{N \times M}$, $\mathbf{B} \in \mathbb{R}^{M \times N}$ and $\mathbf{x} \in \mathbb{R}^N \rightarrow$ this has NM

$$O(N)$$

$\mathbf{x} \cdot \mathbf{y}$ where $\mathbf{x}, \mathbf{y} \in \mathbb{R}^N$ (this is known as the dot product: $\mathbf{x}^T \mathbf{y}$)
Dot product take N multiplications and N-1 additions

$$O(N^2)$$

$\mathbf{x} \otimes \mathbf{y}$ where $\mathbf{x}, \mathbf{y} \in \mathbb{R}^N$ (this is known as the tensor product: $\mathbf{xy}^T$)
this is tensor product meaning = N x N matrix

Problem 2: Use two iterations of Horner's algorithm to calculate $p(x)$ and $p'(x)$ *Horner's Algorithm*

$$p(x) = 3x^4 - x^2 + 2x + 13 \quad \text{for } x = 5$$

$$p(x) = (x - c)Q(x) + R$$

$$R = p(c)$$

$$p'(c) = Q(c)$$

calculate $p(5)$ using synthetic division:

$$\begin{array}{r|rrrrr} 5 & 3 & 0 & -1 & 2 & 13 \\ & \downarrow & 15 & 75 & 375 & 1875 \\ \hline & 3 & 15 & 74 & 377 & 1888 \end{array}$$
$$(x - 5)(3x^3 + 15x^2 + 74x + 377) + 1888 = p(x)$$

calculate $p'(5)$ using synthetic division:

$$p(x) = 3x^3 + 15x^2 + 74x + 377$$
$$\begin{array}{r|rrrr} 5 & 3 & 15 & 74 & 377 \\ & \downarrow & 15 & 166 & 1120 \\ \hline & 3 & 30 & 240 & 1497 \end{array} \quad Q(5) = 1492$$
$$(x - 5)(3x^2 + 30x + 240) + 1492$$
$$p'(5) = Q(5) = 1492$$

$$p(5) = 1873$$

$$p'(5) = 1492$$